

Project ID:

25-26J-035

1. Topic (12 words max)

Energy-Efficient Dual Hydroponic System for Organic/Inorganic Capsicum Chili Farming

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Computer Systems and Network Engineering (CSNE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Hydroponic farming offers a sustainable approach to overcoming food insecurity, rising fertilizer costs, and climate variability challenges, especially in developing countries such as Sri Lanka. Despite its potential, adoption remains constrained by fragmented systems that inadequately integrate energy-efficient communication, adaptive climate control, and intelligent nutrient management for both organic and inorganic crop production under off-grid, solar-powered conditions [1][2][3]. Conventional IoT hydroponic platforms typically focus on isolated tasks like sensor monitoring or automation, without unifying energy forecasting, communication resilience, and holistic environmental control.

A critical limitation lies in communication protocols. Most existing systems rely on energy-intensive TCP/IP-based communication, unsuitable for solar-powered, low-energy environments. Custom lightweight UDP protocols with adaptive transmission frequencies based on solar and battery status, combined with GSM failover, ensure energy-efficient, reliable networking even during Wi-Fi outages. Prioritization of critical sensor data via QoS tagging minimizes energy waste while maintaining responsiveness [5][6]. Real-time network health monitoring with fault tolerance and automated calibration further enhances system reliability.

Intelligent climate control is another underexplored area. Traditional fixed-schedule or threshold-based approaches fail to address dynamic indoor and outdoor conditions. Integrating fuzzy logic-based controls with real-time internal sensor data and external weather APIs allows adaptive regulation of fans, misting, and ventilation tailored for sensitive crops like capsicum.

Predictive fungal risk detection through dew point and leaf wetness estimation enables proactive disease prevention [9][10]. This project uniquely develops dual automated NFT hydroponic systems:

Inorganic NFT System: Featuring real-time monitoring and fuzzy logic control of nutrient dosing, pH balancing, and water flow using multi-sensor data (pH, EC, temperature, humidity, ultrasonic). The system supports sensor calibration, cloud-based visualization, and integrates water reuse with filtration to improve sustainability. Ultrasonic sensors track plant growth and yield progression [4].

Organic NFT System: Utilizing a multi-sensor network including gas and color sensors to classify compost quality via sensor fusion, enabling adaptive nutrient and irrigation automation based on compost state. The system includes an air-gap reservoir for root oxygenation, fuzzy logic control, real-time alerts, cloud connectivity, and manual override options to ensure precise and sustainable organic hydroponic farming [4].

Together, these systems represent a comprehensive solution for precision hydroponics, combining embedded programming, control systems, cloud analytics, and intelligent hardware-software integration. By holistically addressing energy-efficient communication, predictive climate control, nutrient management, and sustainability, this project advances intelligent agriculture suitable for resource-constrained and off-grid environments in developing regions [1][3][4].

References (IEEE Format)

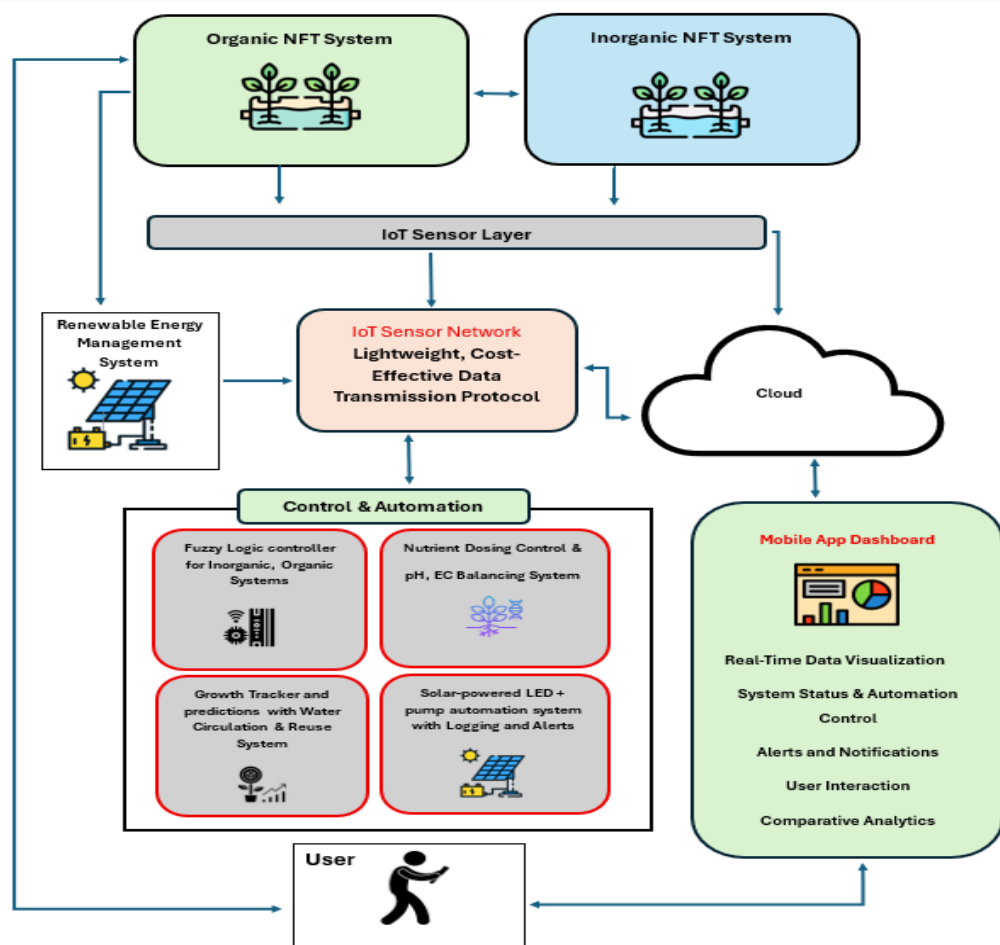
- [1] Bhargava et al., "Sensor Fusion Based Intelligent Hydroponic Farming and Nursing System," IEEE Sensors Journal, 2022.
- [2] Samadder et al., "IoT-Driven Automated Hydroponic System for Climate-Independent Indoor Crop Cultivation," IEEE EDKCON, 2024.
- [3] Mendon et al., "Design and Construction of Automated Hydroponics System with Remote Monitoring and Control," IEEE ICCCNT, 2022.
- [4] R. M. Senthil Priya et al., "IoT Integrated Hydroponic System for Precision Agriculture," IEEE ICPCSN, 2025.
- [5] Satyaprakash Swain et al., "Analytical Performance Comparison of IoT Communication Protocols MQTT and CoAP," IEEE ODICON, 2024.
- [6] Taehyun Kim et al., "A Study on Low Power Wide Area Communication Based Multi-Sensing Technology for Smart Agriculture," IEEE ICUFN, 2024.
- [7] Qi Gao and Dazhi Yang, "Firm Solar Power Forecast through Battery Storage," IEEE CIEEC, 2024.
- [8] Hasyiya Karimah Adli et al., "Forecasting and Analysis of Solar Power Output," IEEE ICICoS, 2021.
- [9] Vanegas-Ayala et al., "Systematic Review of Greenhouse Humidity Prediction & Control with Fuzzy Inference Systems," Wiley, 2022.
- [10] Austria et al., "Development of IoT Smart Greenhouse System for Hydroponic Gardens," arXiv, 2023.

6. Brief description of the nature of the solution including a conceptual diagram

The proposed solution is an IoT-enabled, energy-efficient dual hydroponic system using the nutrient film technique (NFT) to cultivate capsicum chili with high precision. It features two parallel setups organic and inorganic equipped with advanced sensors to monitor environmental, nutrient, and compost conditions such as pH, EC, temperature, humidity, water level, plant height, and compost quality.

A smart climate control zone automatically adjusts fan, mist, and ventilation operations based on internal sensors and external weather forecasts, while also predicting disease risks using dew point and leaf wetness estimation. Sensor data is transmitted through a custom-designed low-power UDP protocol with GSM failover, enabling reliable communication and synchronized data transfer from both systems even in off-grid environments.

All sensor data and system activities are logged and visualized on a cloud-based dashboard that provides real-time monitoring, manual overrides, alerting, and comparative analytics. The entire platform operates using solar power with intelligent scheduling based on solar availability and battery levels, reducing energy waste. This integrated approach offers a smart, reliable, and farmer-friendly hydroponic solution tailored to optimize capsicum chili growth in Sri Lankan conditions.



7. Brief description of specialized domain expertise, knowledge, and data requirements

This project requires specialized multidisciplinary expertise to develop a fully autonomous, energy-aware dual hydroponic system tailored for Sri Lankan conditions. The key specialized domains, knowledge, and data requirements include:

1. IoT Systems & Custom Protocol Design

- In-depth knowledge of interfacing sensors (e.g., pH, EC, temperature, humidity, water level, gas, color, ultrasonic) with microcontrollers (ESP32).
- Design and implementation of lightweight UDP-based communication protocols optimized for low-power, stateless data transmission.
- Integration of Quality of Service (QoS) tagging and GSM failover (using SIM800L) for resilient off-grid and fail-safe data communication.

2. Smart Agriculture & Horticulture Science

- Expertise in NFT hydroponic systems, both organic and inorganic nutrient management, with crop-specific knowledge on capsicum chili physiology and growth stages.
- Ability to calibrate multi-modal sensors and optimize environmental and nutrient parameters for real-time growth tracking and yield prediction.
- Development of compost quality classification using sensor fusion (gas + color sensors) for adaptive organic nutrient dosing.

3. Renewable Energy & Energy Forecasting

- Knowledge of solar power systems, battery charge management, and forecasting solar energy availability to dynamically adapt transmission and operational schedules.
- Skills in power-aware load scheduling for pumps, lighting, climate control devices, and communication modules to maximize uptime and energy efficiency.

4. Automation & Climate Control Systems

- Proficiency in fuzzy logic and rule-based automation for nutrient dosing, irrigation scheduling, ventilation, misting, and fan control.
- Competence in implementing predictive climate controls based on dew point, leaf wetness, and evapotranspiration to mitigate fungal risks and optimize plant health.

5. Network Optimization & Monitoring

- Experience with wireless networking technologies to develop robust systems with dynamic failover switching between Wi-Fi and GSM communication.
- Ability to build real-time network health monitoring modules tracking signal strength, battery status, uptime, and fault tolerance with automated alerts.

6. Cloud Integration

- Competence in cloud data management platforms (e.g., Firebase) and development of interactive dashboards for live monitoring, historical data visualization, and user control interfaces.

Data Requirements:

- Real-time environmental sensor data (pH, EC, humidity, water level)
 - Dataset: <https://github.com/race2infinity/Hydroponics#start-of-content>

- Network performance logs (packet loss, latency, uptime)
 - Dataset: <https://www.kaggle.com/datasets/abdurraziq01/cloud-computing-performance-metrics>
 - Dataset: <https://ieee-dataport.org/documents/internet-protocol-stack-dataset>

- Historical irrigation, nutrient dosing, and lighting logs
 - Dataset: <https://www.kaggle.com/datasets/patricklford/heat-classification-of-chilli-peppers>

- Local weather data for predictive control
 - Source: <https://climateknowledgeportal.worldbank.org/country/sri-lanka/climate-data-historical>

- Energy usage and efficiency logs from IoT devices
 - Dataset: <https://www.kaggle.com/datasets/brunolcarli/iot-energy-consumption-dataset>

- Solar irradiance and panel performance datasets
 - Dataset: <https://ieee-dataport.org/documents/rural-iot-soil-data>

8. Objectives and Novelty

Main Objective

The main objective of this project is to design and implement an energy-efficient, IoT-based dual NFT hydroponic system for capsicum chili cultivation using organic and inorganic nutrient methods. The system enables real-time environmental and nutrient monitoring with intelligent automation optimized for Sri Lankan farming conditions.

Each setup integrates multi-sensor arrays (pH, EC, temperature, humidity, water level, gas, color, ultrasonic) to control nutrient dosing, pH balancing, irrigation, and growth tracking. The organic system uniquely classifies compost quality using sensor fusion to adjust nutrient automation, while the inorganic system features water reuse with filtration and ultrasonic plant height measurement.

A custom low-power UDP communication protocol with QoS and GSM failover ensures reliable data transmission in solar-powered, off-grid environments. Energy management modules dynamically schedule device operation based on solar and battery status.

Fuzzy logic-based climate control uses internal sensors and weather data to regulate ventilation, misting, and fans, with predictive fungal risk detection via dew point and leaf wetness. Data streams to a cloud dashboard offering real-time visualization, alerts, and comparative analytics, empowering farmers with scalable, automated hydroponic farming.

This integrated platform provides a sustainable, cost-effective solution tailored for precision agriculture in resource-limited settings.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Galappaththi D. S IT22355782	Design a low-power adaptive network for dual hydroponics that adjusts data based on tank status and Solar energy	Define lightweight, stateless UDP packet structure for sensor data transmission Design and implement the hardware system for networking and power management Monitor solar/battery levels to dynamically adapt transmission frequency for maximum energy efficiency Integrate GSM failover using SIM800L for reliable communication during Wi-Fi outages Implement QoS tagging to prioritize critical sensor data while minimizing energy waste Synchronize data transmission from organic and inorganic tanks	Custom UDP protocol optimized for low-power, stateless data transmission, minimizing energy use compared to TCP. Adaptive scheduling based on solar/battery forecasting to maximize uptime and efficiency. QoS-based prioritization ensures critical sensor data is transmitted reliably. Dual failover system (Wi-Fi + GSM) guarantees continuous communication even during outages. Real-time network health monitoring supports proactive maintenance and system reliability.

		Measure and analyze energy savings compared to traditional TCP communication Automate calibration and alerts for network fault tolerance Build a network health monitoring module (signal strength, battery, uptime).	
Kariyawasam K.S.T.S. A IT22617996	Build an automated climate zone that adapts to weather and predicts fungal risks for healthy capsicum growth.	Define optimal temperature, humidity, and leaf wetness ranges for each capsicum growth stage. Set up sensors (temperature, humidity, CO₂, optional leaf wetness) and connect to ESP32. Control fans, misting, and ventilation based on real-time sensor data using fuzzy logic rules. Fetch external weather via APIs and compare with indoor data for predictive control.	Real-time climate control using both internal sensor data and external weather inputs. Predictive fungal risk detection via dew point and leaf wetness estimation. Automated fan/mist control tailored to capsicum growth needs. Early alerts for climate-triggered plant health threats — unique to this zone.

		Estimate dew point and leaf wetness to detect fungal risk and auto-trigger preventive actions. Generate daily logs and graphs of climate activity and disease alerts.	
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Weerasekara P.A.S. S IT22258762	Design and implement an automated inorganic NFT hydroponic system with real-time monitoring and intelligent nutrient management to optimize capsicum chili growth.	Construct the complete inorganic NFT setup including nutrient tank, grow channels, and water circulation system. Install sensors (pH, EC, water level, temperature, humidity, ultrasonic/ToF) and integrate with ESP32 for real-time data collection and control. Implement fuzzy logic automation on ESP32 to control nutrient dosing, pH balancing, and water flow using multiple pumps. Integrate water reuse with basic filtration to improve sustainability and reduce waste. Develop cloud connectivity and dashboard to visualize sensor data, nutrient levels, and plant growth trends in real-time. Enable semi-automated sensor calibration and use ultrasonic/ToF sensor to monitor plant height and yield progression.	Real-time fuzzy logic control for nutrient and pH adjustment tailored to inorganic systems. Sensor based growth tracking is integrated with cloud analytics. Water reuse and recycling mechanism for nutrient solution with in-line filtration and sterilization Semi-automated sensor calibration for reliable long-term monitoring. Nutrient recycling system with filtration
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Dissanayaka D.M.I. P IT22167682	Design and implement an automated organic NFT hydroponic system with real-time monitoring and intelligent nutrient management to optimize capsicum chili growth.	Build an NFT hydroponic system using organic compost tea and include an air-gap reservoir to improve root oxygenation. Install and calibrate sensors (pH, EC, temperature, humidity, ultrasonic water level, TDS, gas, and color) to monitor environmental and compost conditions. Develop a compost classification model to identify Fresh / Ready / Over-fermented compost using gas and color sensor data Implement automation using fuzzy logic to control nutrient dosing, irrigation, and pH balancing based on sensor inputs and compost state Control dosing pumps to automatically adjust water and nutrient levels according to system needs. Create a dashboard to display real-time sensor data, compost status, control logs, and add manual override and alert features.	Semi-automated organic hydroponic system with advanced sensing and control. Uses sensor fusion (gas + color) to classify compost quality in real-time. First to apply compost quality-based nutrient automation in an NFT system. Supports cloud-based monitoring, real-time alerts, and manual override. Enables sustainable smart farming with organic nutrients, minimal human input, and efficient resource use.
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Galappaththi D. S IT22355782	This component involves the design and implementation of a custom, lightweight UDP-based communication protocol, optimized for energy efficiency in solar-powered, embedded IoT systems. It aligns directly with the Computer Systems and Network Engineering (CSNE) specialization by integrating core concepts such as embedded systems (ESP32, sensor interfacing), wireless network protocol design (UDP vs TCP analysis), Quality of Service (QoS) handling, and energy-aware scheduling
Kariyawasam K.S.T.S. A IT22617996	Develop an adaptive climate control zone for capsicum chili that uses sensors (temperature, humidity, CO ₂ , leaf wetness) and fuzzy logic to regulate ventilation, misting, and fans. It integrates real-time internal data with external weather forecasts to predict fungal risks through dew point and leaf wetness estimation. This system automates preventive actions, supports low-energy operation, and provides daily climate and disease alerts for optimized, sustainable crop health management.
Weerasekara P.A.S. S IT22258762	This part involves the development of an automated inorganic NFT hydroponic system. It includes sensor integration (pH, EC, temperature, humidity), fuzzy logic for nutrient and pH control, and real-time cloud monitoring. The system uses ESP32 microcontrollers and supports semi-automated sensor calibration. Additional features include a water reuse module and ultrasonic sensor-based plant growth tracking. This aligns with CSNE through embedded programming, control systems, cloud-connected monitoring, and intelligent hardware-software integration.
Dissanayaka D.M.I. P IT22167682	This component covers the design of an IoT-based organic NFT hydroponic system with compost validation and adaptive automation. It involves a multi-sensor network to track nutrient quality and environmental conditions, and uses fuzzy logic to control pH, dosing, and irrigation. The member implements a compost quality classification system using sensor fusion and supports cloud connectivity for monitoring and control. This aligns with CSNE through embedded IoT integration, real-time system feedback, and sustainable automation techniques.

10. Supervisor's details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Shashika	Lokuliyana	
Co-Supervisor	Mr.	Dinith	Primal	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

1) Update the objectives, tasks & novelty component.

Staff Member's Name	Signature
Shashika Lokuliyana	

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.